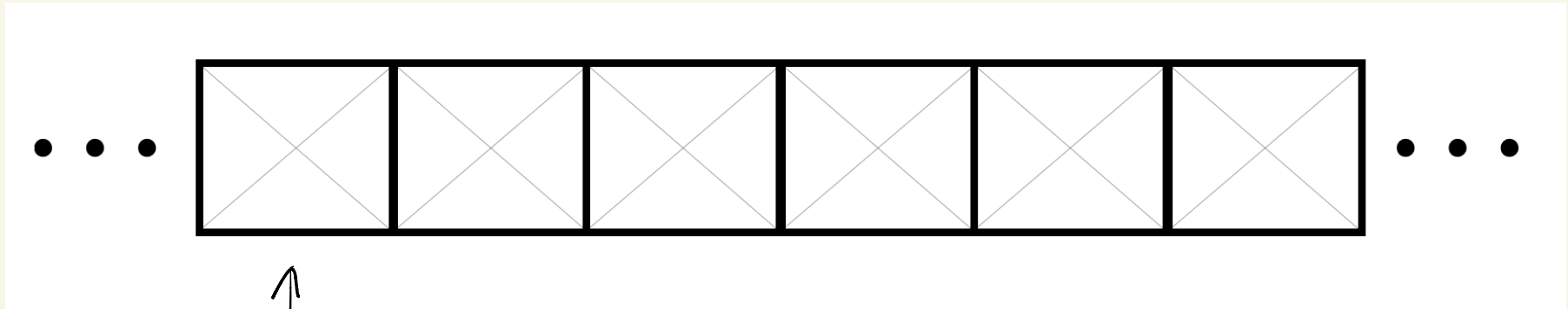


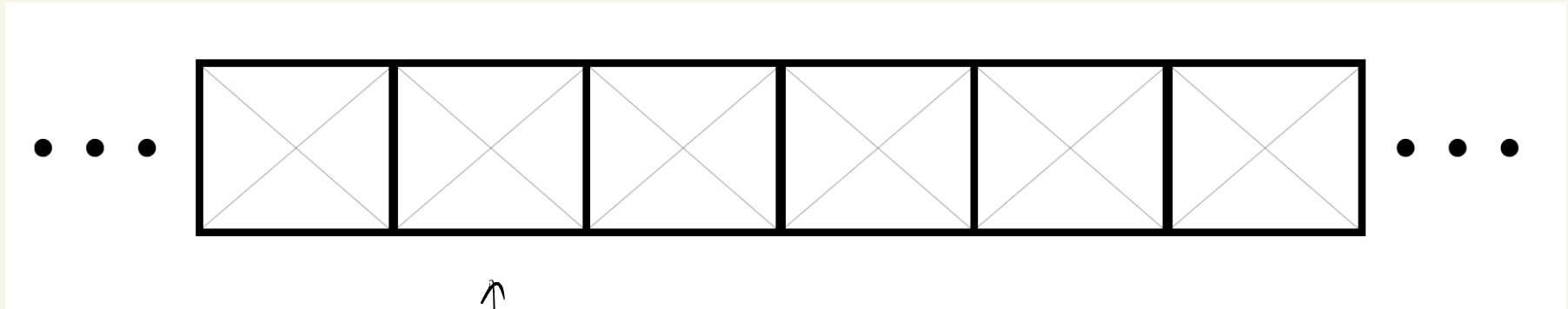
Braven's Algorithm



unstructured database . find 56

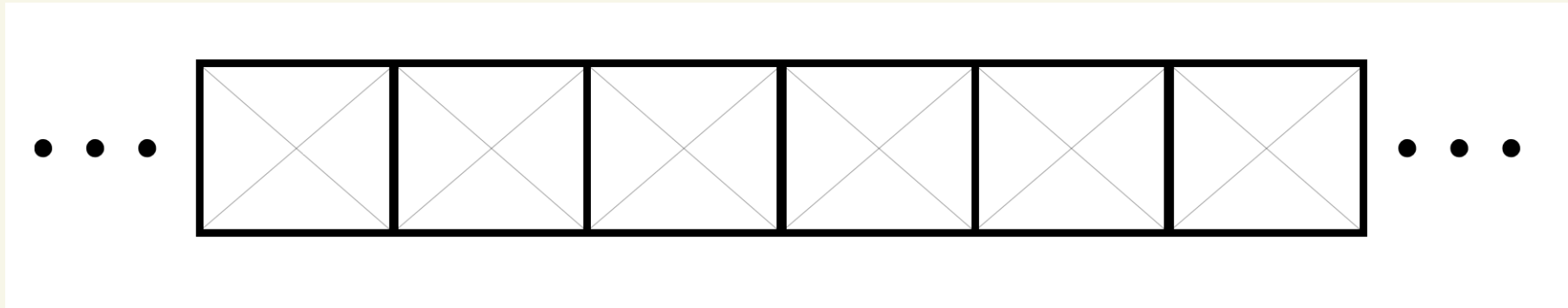


Are you
56?
No, OK!



Are you
56?

$\longleftrightarrow N \longrightarrow$

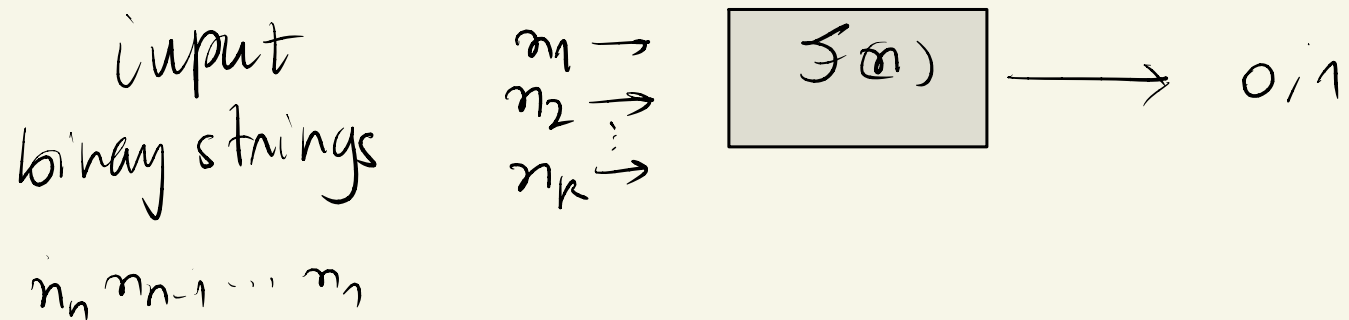


classically $\Rightarrow O(N)$

Grover $\Rightarrow O(\sqrt{N})$

Let's imagine we have a Boolean function $\mathcal{F}$.

Boolean functions $\mathcal{F}: \{0,1\}^n \mapsto \{0,1\}$



⊗ Guaranteed that a single binary string satisfies $\mathcal{F}$.

How many calls (queries) to $\mathcal{F}$ do we need to find x ?

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to find x ?

for n -bit bitstrings there are $6(2^n)$
possible bitstrings

classically $\Rightarrow 6(2^n)$

Grover $\Rightarrow 6(\sqrt{2^n})$

Satisfiability problems

E.g. scheduling problem

Determine when a meeting can occur
given constraints on people's availability

$\{\text{Monday}, \dots, \text{Friday}\} \mapsto \{n_1, n_2, \dots, n_5\}$

João $\rightarrow$ Monday, Tuesday, Friday

Carlos $\rightarrow$ not Friday

Diana $\rightarrow$ Monday or Wednesday

$\vdots$

$$\mathcal{F}(n_1, n_2, n_3, n_4, n_5) = (n_1 \vee n_2 \vee n_5) \quad \text{João}$$

$$\wedge$$

$$(\neg n_5) \quad \text{Carlos}$$

$$\wedge$$

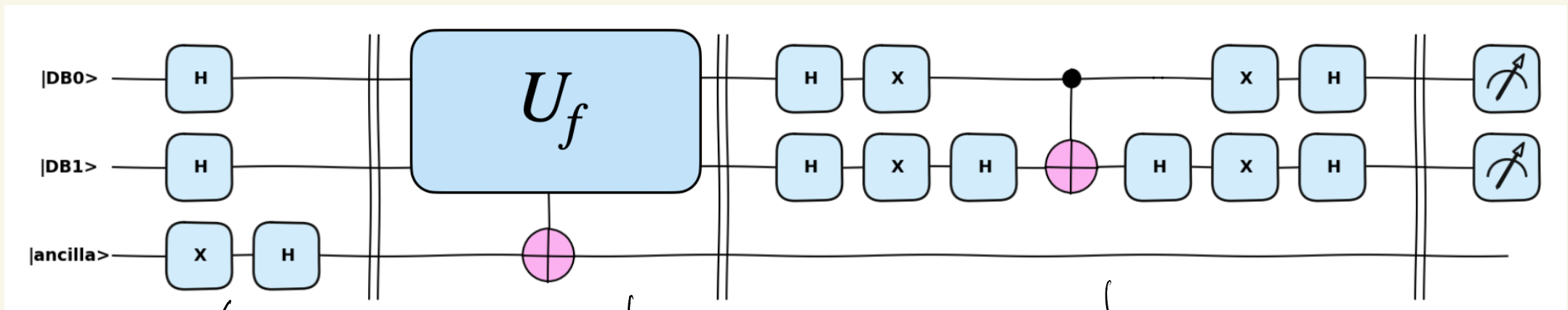
$$(n_1 \vee n_3) \quad \text{Diana}$$

$$\wedge$$

$$\dots$$

Find $(n_1, n_2, n_3, n_4, n_5)$ that satisfies $\mathcal{F}$.

Grover's Algorithm



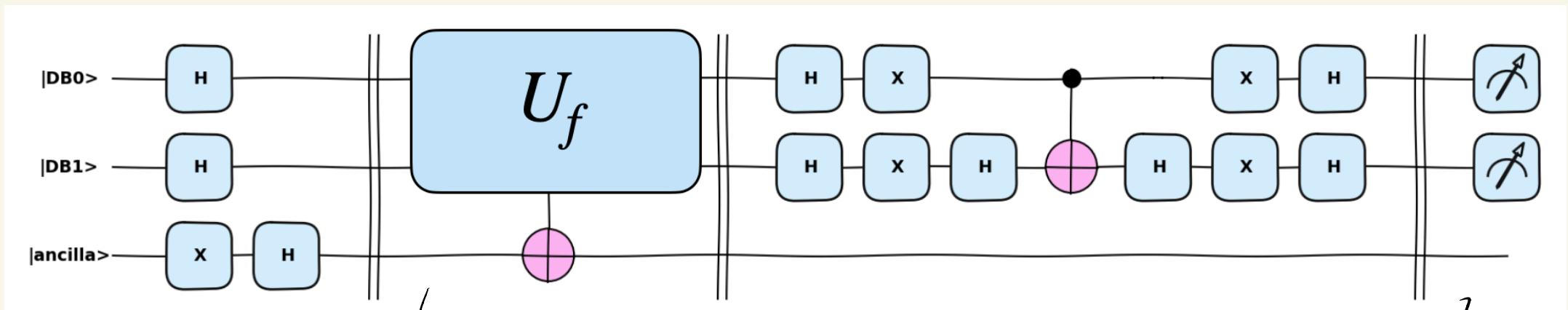
Database
Superposition
+
ancilla $1 \rightarrow$

ORacle
function evaluates
every possible
input

Diffusion operator

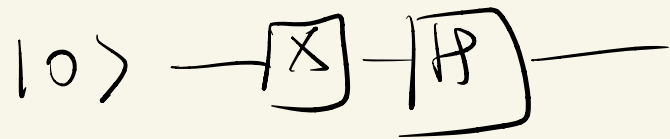
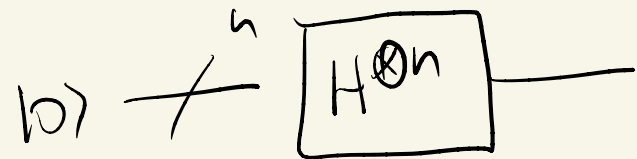
Amplifies probability
of measuring
solution

Grover's Algorithm



Repeat $O(\sqrt{N})$ times

Initial states



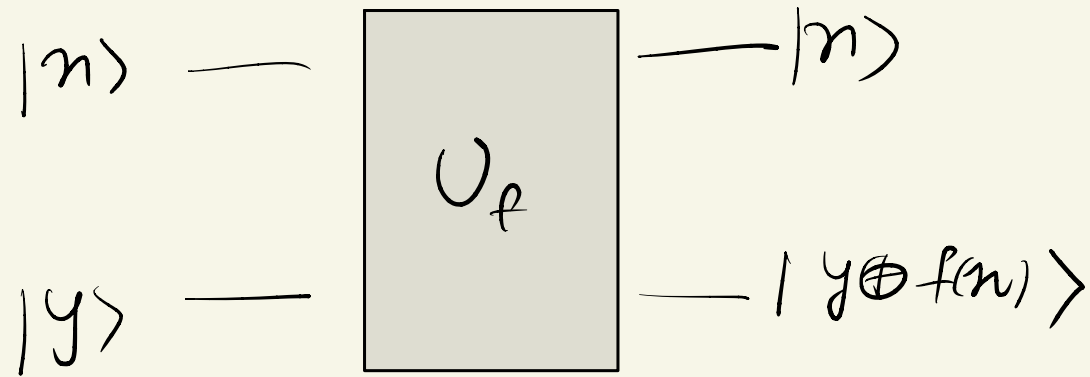
$$(H^{\otimes n} \otimes HX) (|0\rangle^{\otimes n} \otimes |0\rangle)$$

$$= \frac{1}{\sqrt{2^n}} \sum_{i=0}^{2^n-1} |x_i\rangle \quad |-\rangle$$

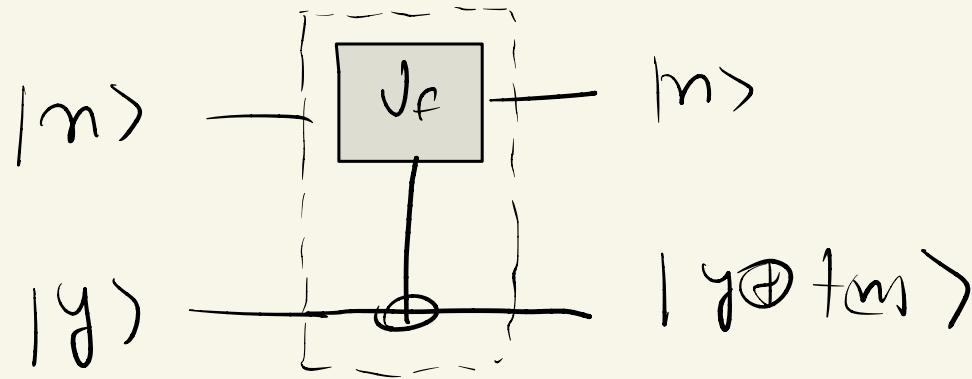
ORACLE

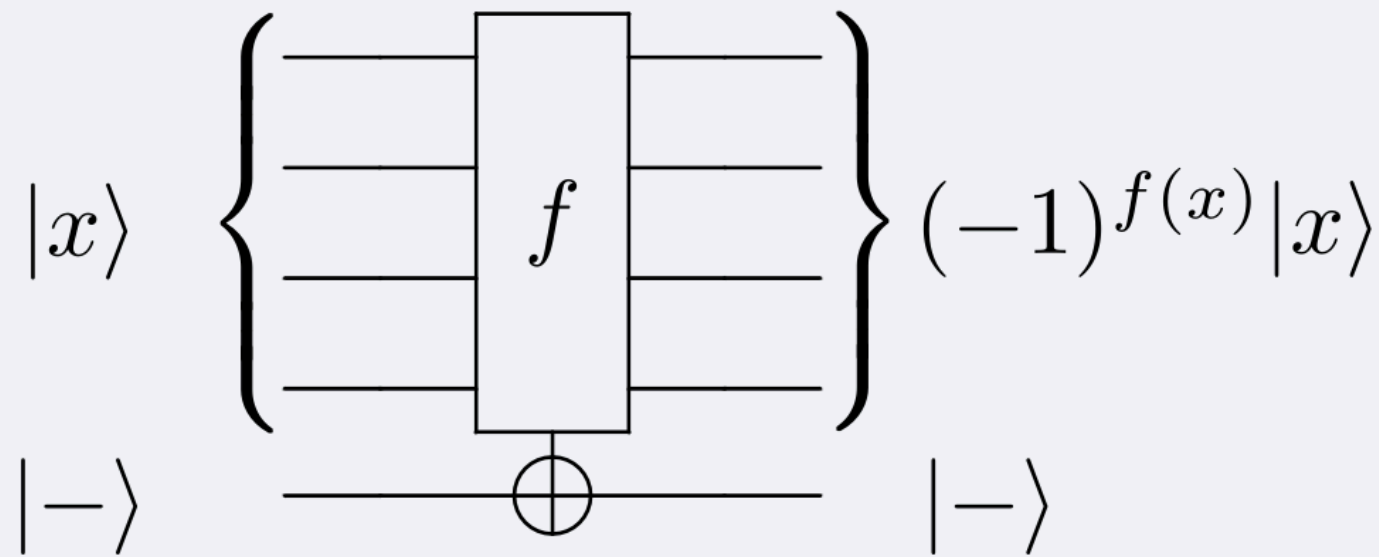


Quantum query gates



U_f makes bitflip on $|y\rangle$:

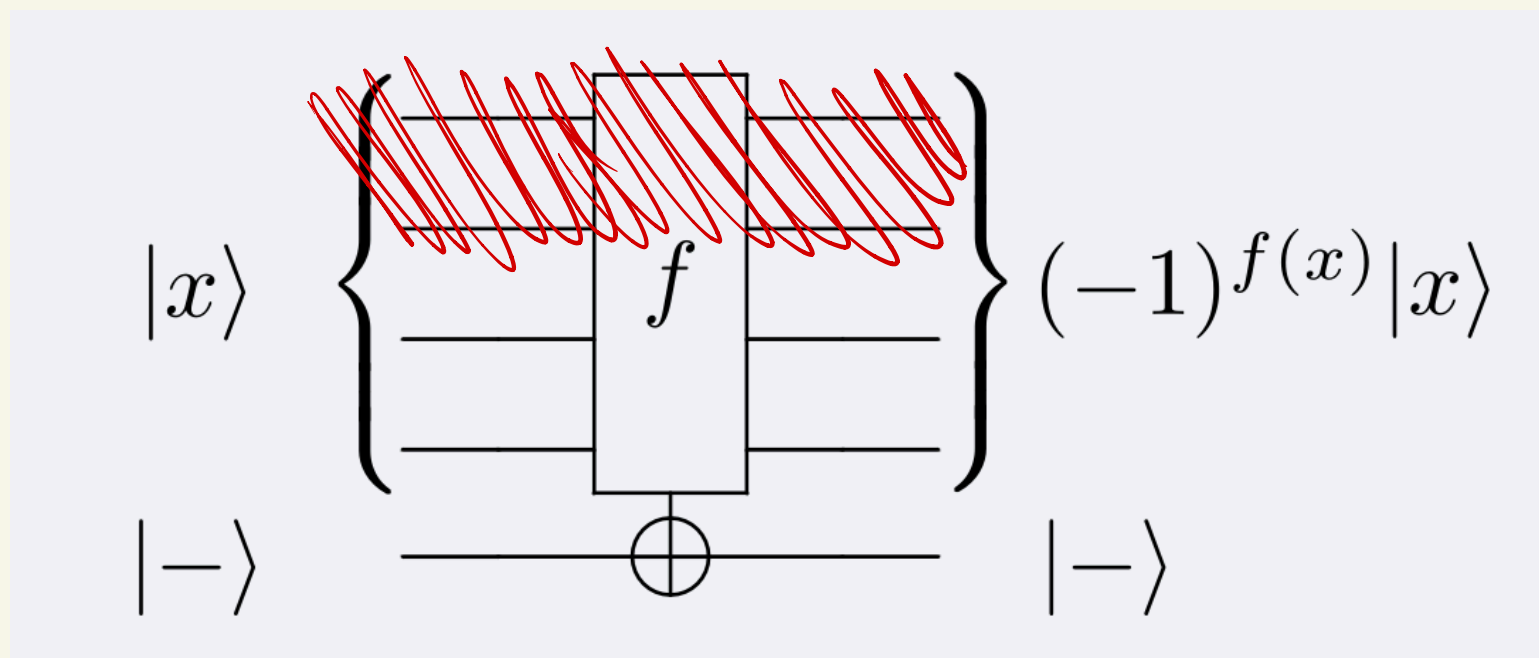




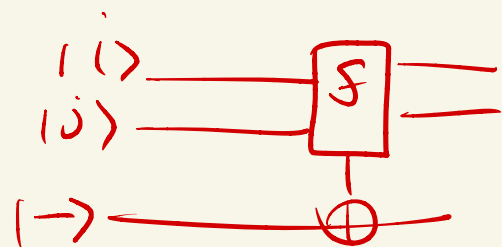
$|x\rangle$ satisfies $\mathcal{F}$. Then, from phase kickback,

$$\begin{aligned}
 \mathcal{F} |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) &= |x\rangle |0 \oplus f(x)\rangle - |x\rangle |1 \oplus f(x)\rangle \\
 &= |x\rangle |1\rangle - |x\rangle |0\rangle \\
 &= -|x\rangle |-\rangle = (-1)^{f(x)} |x\rangle |-\rangle
 \end{aligned}$$

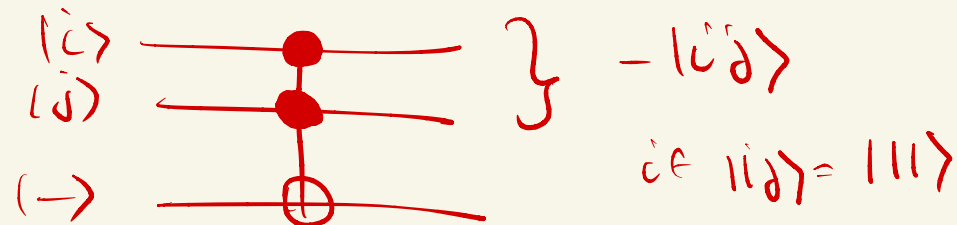
e.g.



2 qubits only : database with 4 elements
Imagine $|11\rangle$ is the solution



=



7me. Toffoli

$$U_f \left(\frac{1}{\sqrt{2^n}} \sum_{j=0}^{2^n-1} |x_j\rangle \right) \rightarrow$$

Let's assume a single solution $|w\rangle$

$$U_f \left(\frac{2^{n-1}}{\sqrt{2^n}} \sum_{n \neq w} |n\rangle + \frac{1}{\sqrt{2^n}} |w\rangle \right) \rightarrow$$

⊗ If we measure at this stage then
we get $|w\rangle$ with $\frac{1}{2^n}$ probability

$$U_f \left(\frac{1}{\sqrt{2^n}} \sum_{j=0}^{2^n-1} |x_j\rangle \right) \rightarrow$$

Let's assume a single solution $|w\rangle$

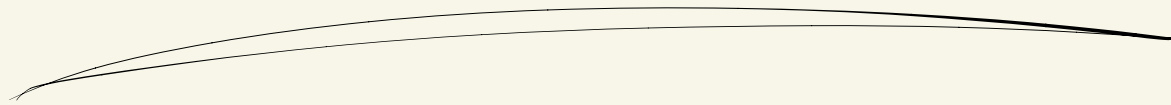
$$U_f \left(\frac{2^{n-1}}{\sqrt{2^n}} \sum_{n \neq w} |n\rangle + \frac{1}{\sqrt{2^n}} |w\rangle \right) \rightarrow$$

invert the phase of solution $|w\rangle$

⊗ If we measure at this stage then
we get $|w\rangle$ with $\frac{1}{2^n}$ probability

→ Diffusion operation amplifies probability of $|w\rangle$

Diffusion operator

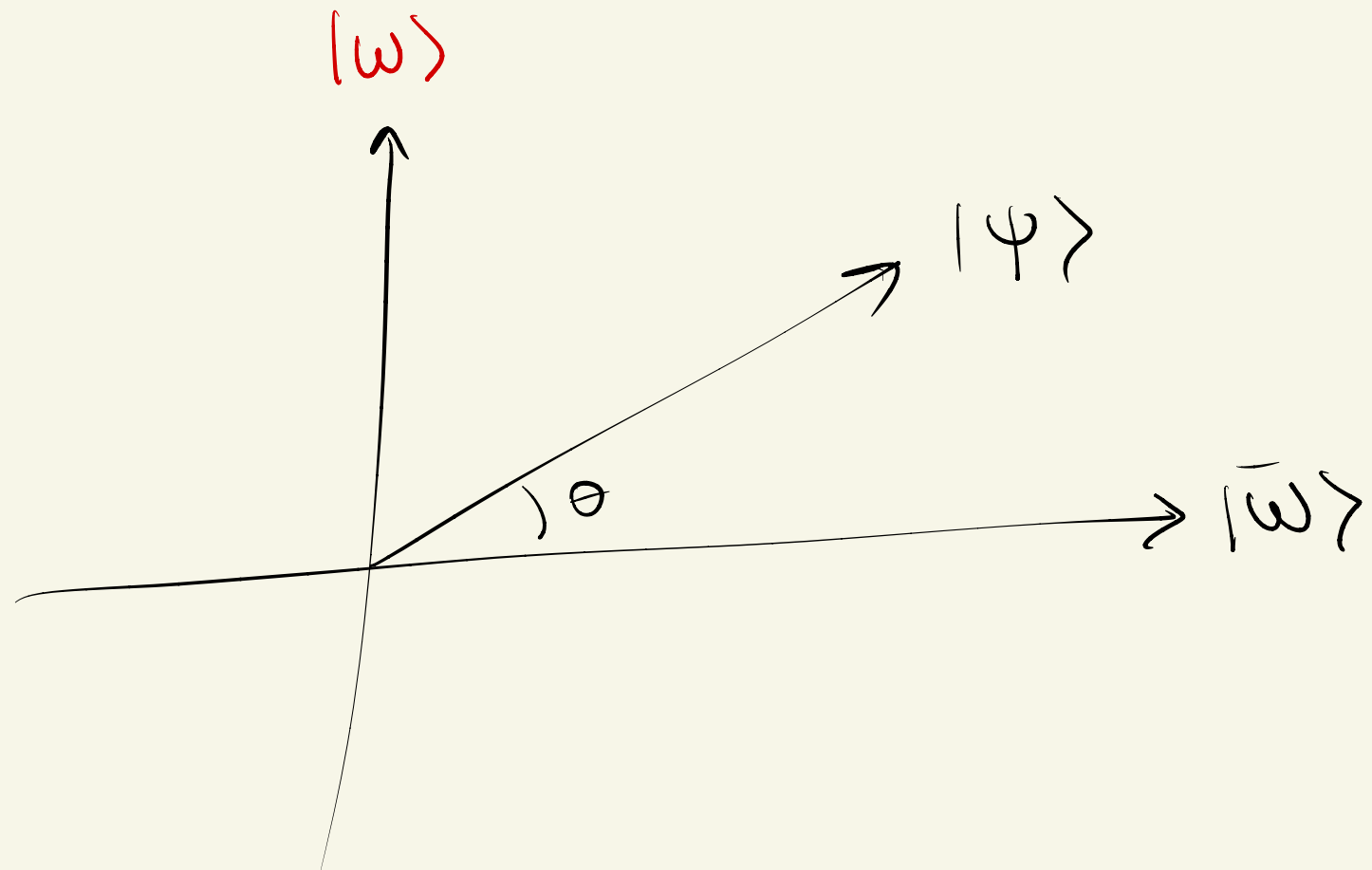


Before oracle (ignoring ancilla)

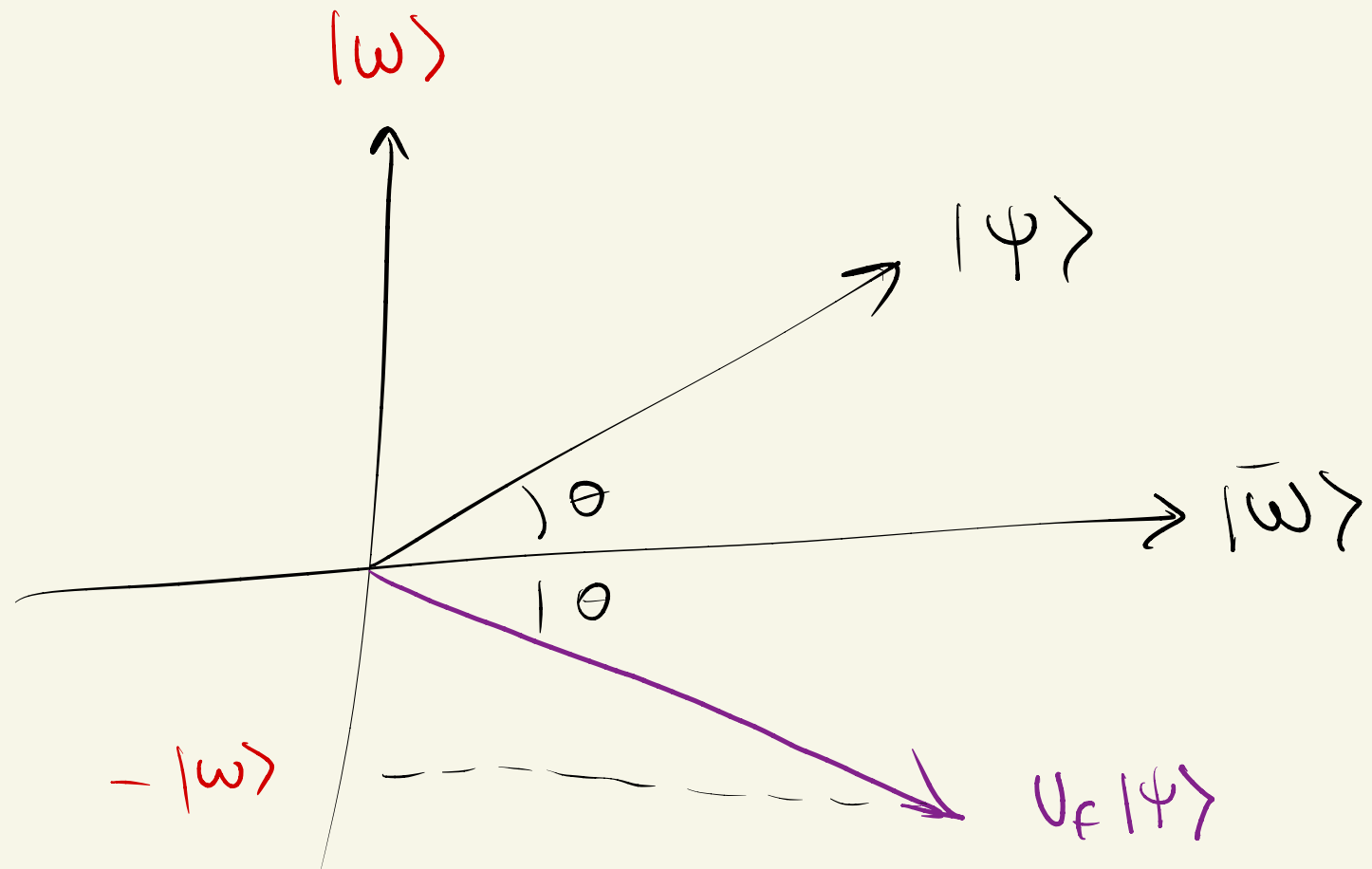
$$\frac{2^n - 1}{\sqrt{2^n}} \sum_{n \neq w} |n\rangle + \frac{1}{\sqrt{2^n}} |w\rangle = |\psi\rangle$$

$$= \frac{2^n - 1}{\sqrt{2^n}} |\bar{w}\rangle + \frac{1}{\sqrt{2^n}} |w\rangle$$

$$= \cos(\theta) |\bar{w}\rangle + \sin(\theta) |w\rangle$$

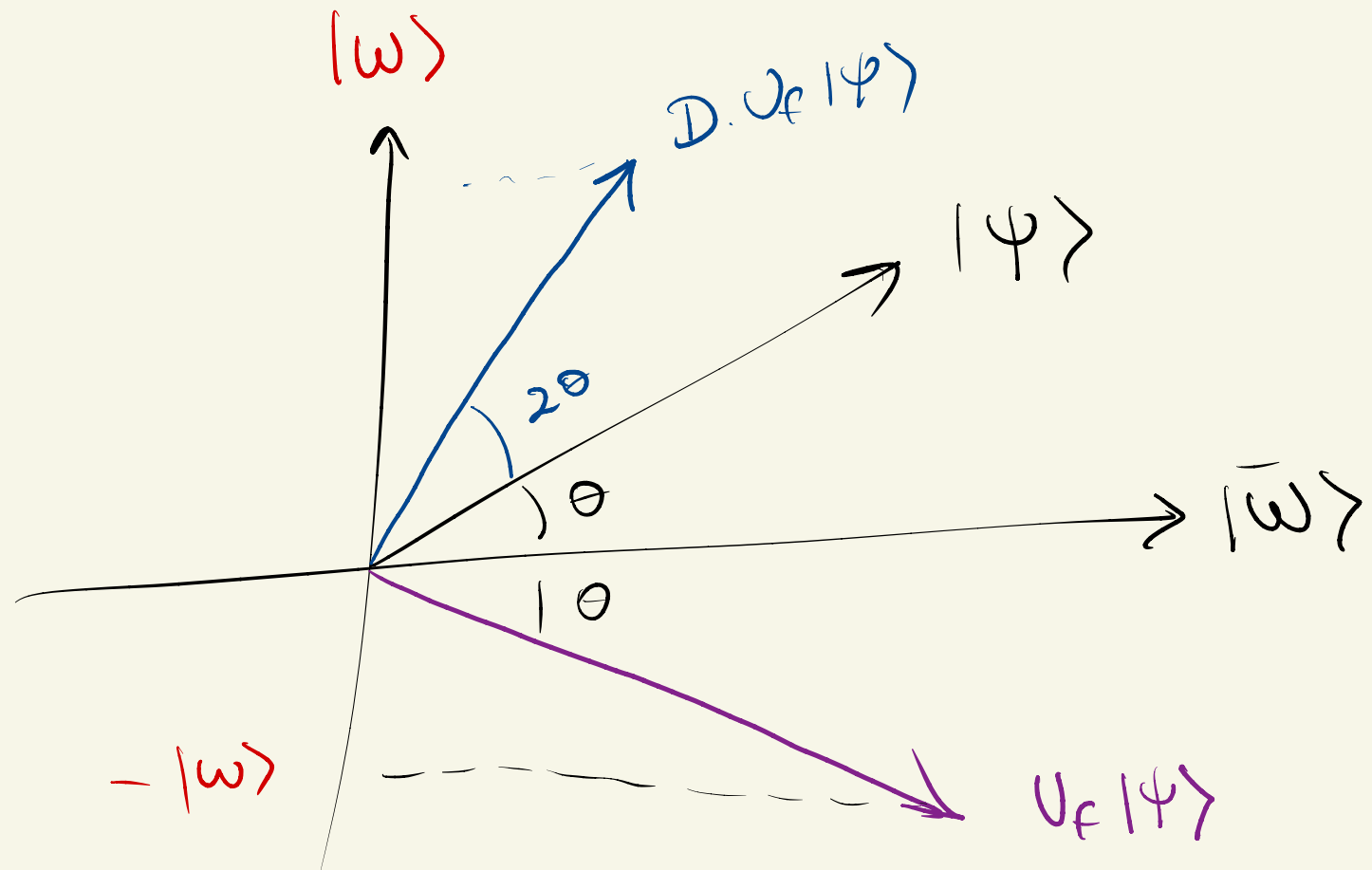


Effect of the oracle on $|\psi\rangle$?



what if we reflected over $|\psi\rangle$?

Probability would increase with
an angle of 3θ !



$$D = 2|\psi\rangle\langle\psi| - I$$

$$D = 2|\psi\rangle\langle\psi| - I$$

$$= 2 H^{\otimes n} |0\rangle\langle 0| H^{\otimes n} - I$$

$$= 2 H^{\otimes n} |0\rangle\langle 0| H^{\otimes n} - H^{\otimes n} H^{\otimes n}$$

$$= H^{\otimes n} (2|0\rangle\langle 0| - I) H^{\otimes n}$$

$$|0\rangle\langle 0| = \begin{pmatrix} 1 & & \\ & 0 & \\ & 0 & \ddots \\ & \ddots & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 \end{pmatrix}$$

$$2|0 \times 0\rangle - I$$

$$= \begin{pmatrix} 1 & & & & & \\ & -1 & & & & \\ & & -1 & & & \\ & & & -1 & & \\ & 0 & & & \ddots & \\ & & & & & -1 \end{pmatrix}$$



-1 phase in every state
except $|0\rangle$.

$$2|0\rangle\langle 0| - I$$

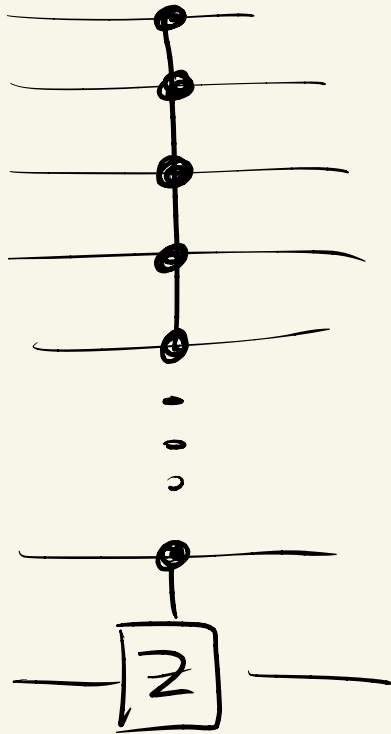
$$= - \begin{pmatrix} -1 & & & & & \\ & +1 & & & & \\ & & +1 & & & \\ & & & +1 & & \\ & 0 & & & \ddots & \\ & & & & & +1 \end{pmatrix}$$



-1 phase in state $|0\rangle$

what gate implements this?

multi controlled z gate



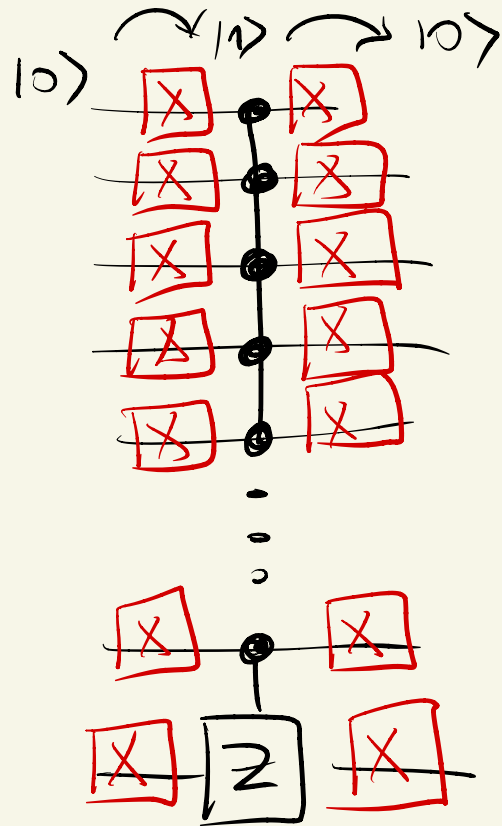
Applies phase (-1)

to state $|111\dots 1\rangle$

but we want

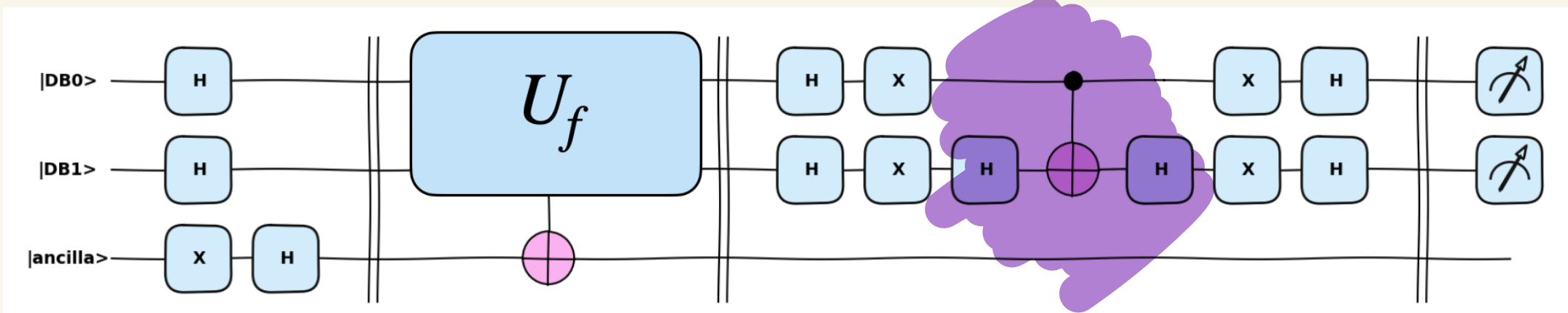
$|000\dots 0\rangle$

multi controlled z gate



Applies phase (-1)

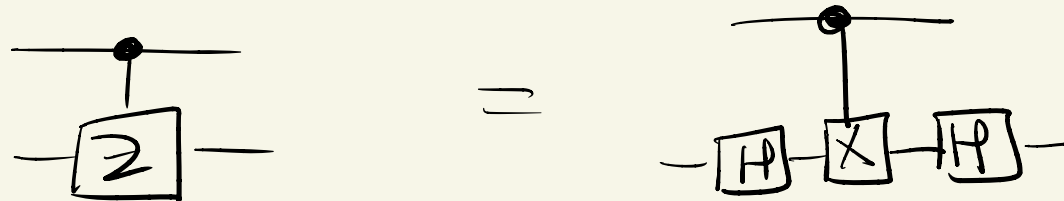
to state $|000\dots 0\rangle$



?

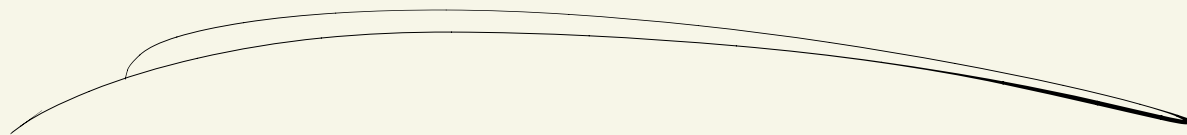
$$Z = H \times H$$

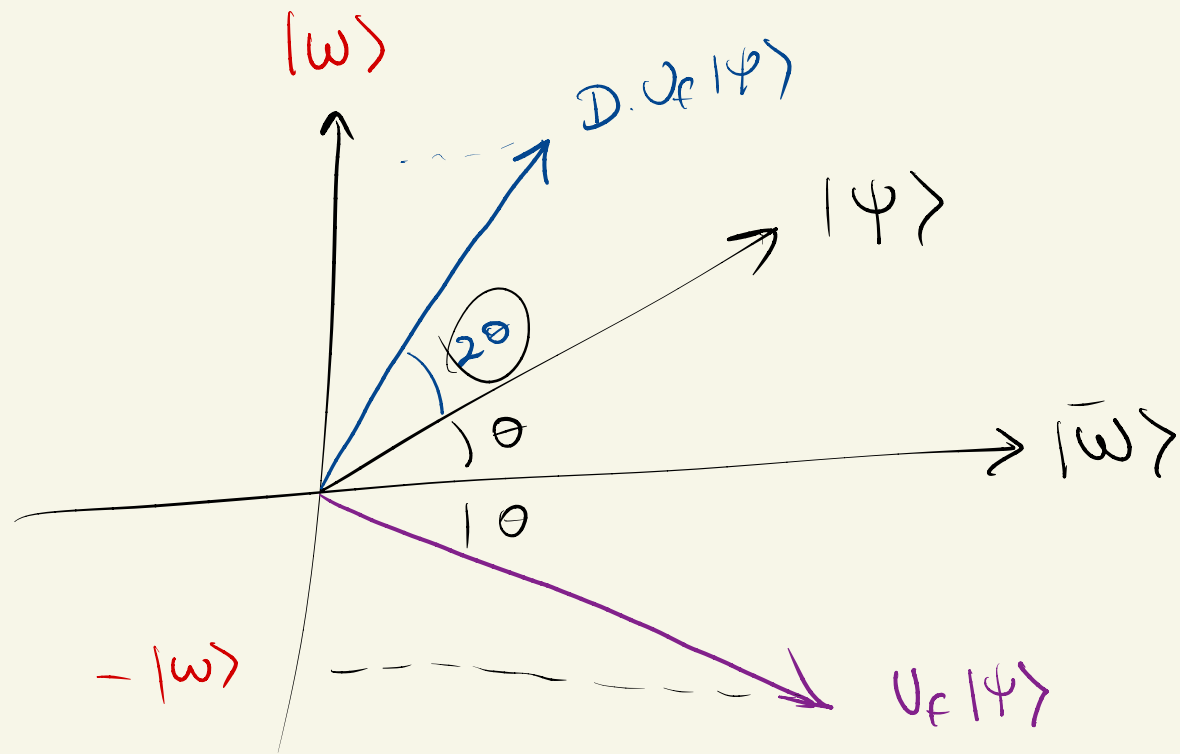
Extends to controlled gates



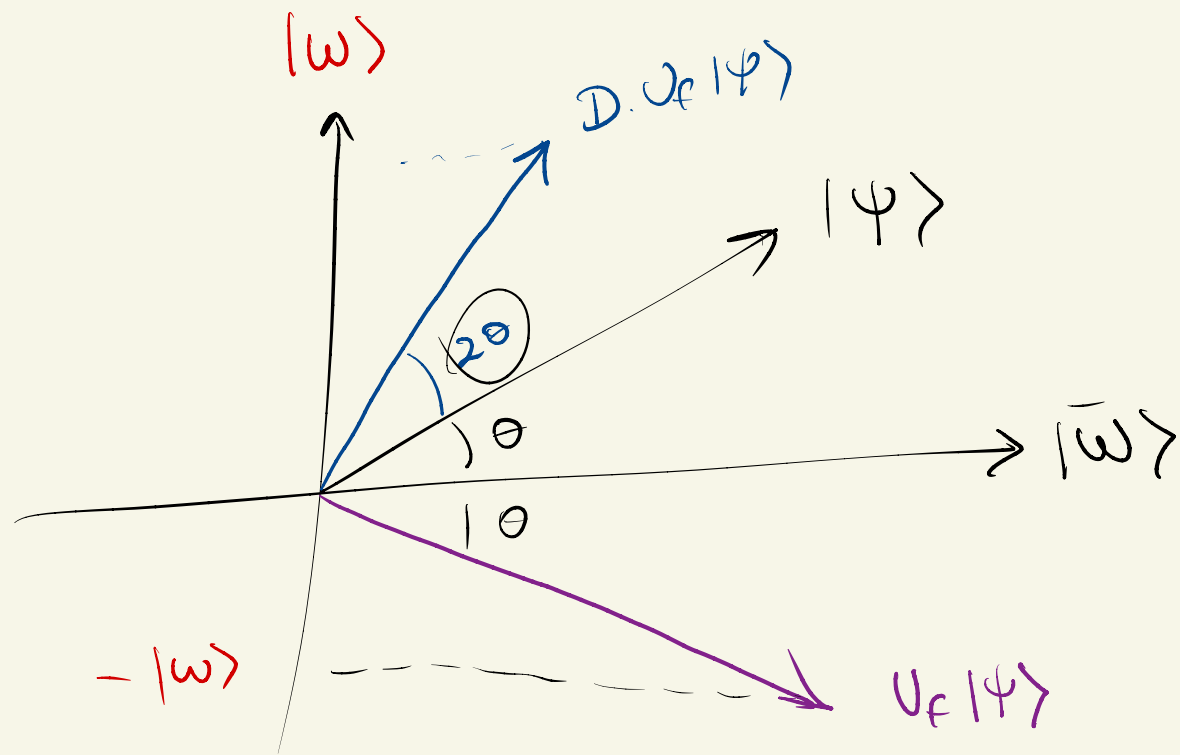
How many times
should we apply

$U_f + D$?



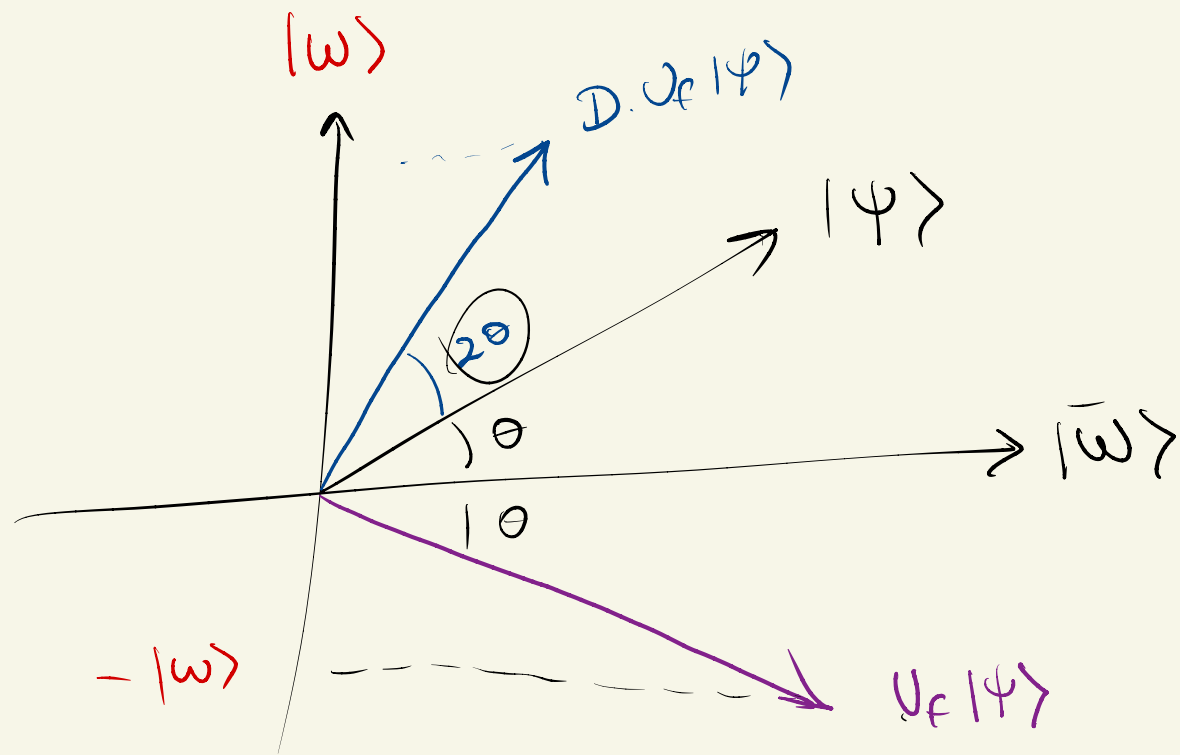


$$D.U_f|\psi\rangle = \cos(\theta + 2\theta)|\bar{w}\rangle + \sin(\theta + 2\theta)|w\rangle$$



$$D.U_f|\psi\rangle = \cos(\theta + 2\theta)|\bar{w}\rangle + \sin(\theta + 2\theta)|w\rangle$$

So for j iterations $(D.U_f)^j|\psi\rangle$



So for j iterations $(D.U_f)^j |\psi\rangle$

$$D.U_f |\psi\rangle = \cos((2j+1)\theta) |\bar{w}\rangle + \sin((2j+1)\theta) |\omega\rangle$$

Each application of D adds 2θ .

for j iterations, the goal is:

$$\sin^2((2j+1)\theta) \approx 1$$



$$\sin^2(\theta) = \frac{1}{2^n}$$

$$\theta = \arcsin\left(\frac{1}{\sqrt{2^n}}\right)$$

$$\arcsin(1) = \frac{\pi}{2}$$

$$J \approx \frac{\pi}{4} \frac{1}{\arcsin\left(\frac{1}{\sqrt{N}}\right)} - \frac{1}{2}$$

$$\arcsin(x) \approx x \text{ for } |x| \ll 1 \quad N \rightarrow \infty$$

Asymptotically, $J \in O\left(\sqrt{2^N}\right)$

$\sqrt{2^N}$ calls to the "database" or function to find solution!

Questions:

→ what if S has more than one solution?

→ what if we do not know the number of solutions?